

Chapter 47

Editing, Adding, and Copying Effects and Filters

This chapter covers how to browse for and apply effects to clips in the Timeline, how to copy them from clip to clip, how to remove them, and how to edit them in the Inspector once they've been added.

For more information about the specific Resolve FX that are available, see “Resolve FX.”

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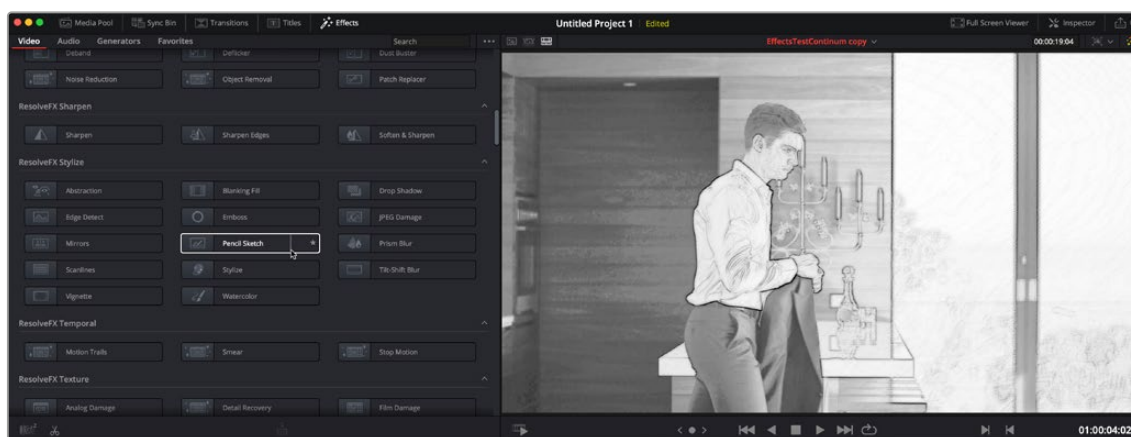
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Using the Effects Library

All effects that you can add to your edit, including filters, transitions, titles, and generators, are found in the Effects panel. The Effects panel shows a hierarchical list of all of the different transitions, title effects, generators, and filters that are available, sorted by category.

To preview a video effect before placing it on a clip, ensure that “Hover Scrub Preview” is checked in the Effects option menu, then simply hover your pointer over any thumbnail in the Effects tab and move it across the thumbnail. The effect will preview in the Viewer using its default parameters, and scrub through the clip that is selected in the Timeline. If no clip is selected then it will use the clip currently under the playhead.

To activate a specific video effect on a clip, simply drag the thumbnail of the selected effect to a clip on the Timeline. You can also double click the thumbnail, or drag an effect directly into the Inspector to apply the effect to the selected clip. To adjust the effect’s parameters, open the Effects tab in the Inspector.



Scrubbing over an Effect Thumbnail previews that effect in the Viewer.

You can search for a particular effect by clicking on the magnifying glass icon at the top of the Effects Library. A drop-down menu just to the right of it lets you choose whether to search the entire FX library (All Folders) or just the current folder.

You can search for effects and transitions using Category Names as a search term. For example, searching for “Stylize” will bring up all the Resolve FX under the Stylize section.

The Toolbox

All of the video and audio transitions, titles, and generators that ship along with DaVinci Resolve:

Toolbox

Exposes all transitions, titles, generators, and effects at once.

Video Transitions: Contains all of the built-in transitions that are available from DaVinci, organized by category. At the bottom of the list, a User category shows presets that you’ve saved. You can drag any video transition to any edit point in the Timeline that has overlapping clip handles to add it to your edit; you have the option to drag the transition so that it ends on, is centered on, or starts on the edit point.

For more information, see *Chapter 48, “Using Transitions.”*

Audio Transitions: Contains audio transitions for creating crossfades.

Titles: Titles can be edited into the Timeline like any other clip. Once edited into the Timeline, you can edit the title text and position directly in the Timeline Viewer, or you can access its controls in the Inspector for further customization.

Generators: Generators can also be edited into the Timeline like any other clip. Selecting a generator and opening the Inspector lets you access its controls for further customization. You can also choose a standard duration for generators to appear within the Editing panel of the User Preferences.

Effects: Effects are essentially placeholders in the Timeline that allow for more specialized compositing in Fusion, or that let you modify the underlying tracks with an adjustment clip.

Open FX

DaVinci Resolve supports the use of third-party Open FX filters, transitions, and generators in the Edit page. Once you install these effects on your workstation, they appear in this section of the Effects Library, organized by type and group depending on the metadata within each effect.

Open FX

Exposes all Resolve FX and third-party Open FX installed on your workstation at once.

Filters: Contains the Resolve FX filters that ship with DaVinci Resolve, as well as any third-party OFX plugins you've installed on your workstation. Filters can be dragged onto video clips to apply an effect to that clip. Once applied, filters can be edited and customized by opening the Open FX panel of the Inspector.

Generators: Contains any third-party OFX generators you have installed on your workstation. Can be edited into the Timeline just like the native generators that ship with DaVinci Resolve, but they also expose an Open FX panel next to the Transition panel in the Inspector, where you can customize settings that are unique to that transition.

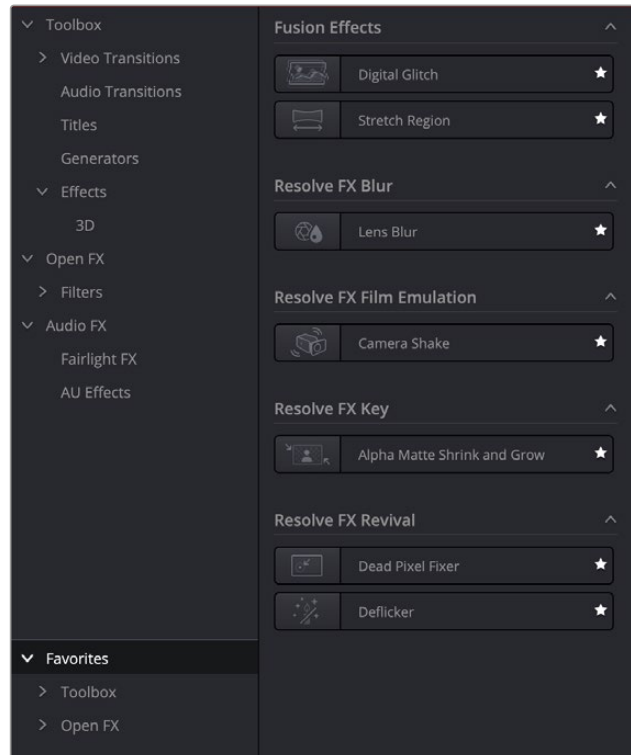
Transitions: Contains any third-party OFX transitions you have installed on your workstation. OFX transitions can be used similarly to any other transition, but they also expose an Open FX panel next to the Transition panel in the Inspector, where you can customize settings that are unique to that transition.

Audio FX

On all platforms, DaVinci Resolve supports a set of built-in Fairlight FX, a native audio plugin format that makes various audio tools and effects available on macOS, Windows, and Linux. On macOS and Windows, DaVinci Resolve supports the use of third-party VST audio plugins, which includes most of the professional third-party tools and effects used in the audio postproduction industry. On macOS, DaVinci Resolve supports Audio Unit (AU) audio plugins. Once you install these effects on your workstation, they appear in this panel of the Effects Library. Audio plugins let you apply effects to audio clips or entire tracks' worth of audio, to add creative qualities such as echo or reverb, or to take care of mastering issues using noise reduction, compression, or EQ.

Effects Library Favorites

You can click on the far right of any transition, title, or generator flag that effect with a star as a favorite effect. When you do so, the favorited effects appear in a separate Favorites area at the bottom of the Effects Library Bin list. Favorited effects will be automatically categorized in the list.



Stars indicate a flagged favorite effect; all favorites are filtered and automatically categorized.

Converting Fusion Compositions to Edit Effects

If you have created a Fusion Composition that you would like to use across a number of projects, you can now save it to the Effects toolbox in the Edit page. This allows easy access to the effect from the Edit page, and simple application to video clips in the Timeline.

To convert a Fusion composition to an Edit effect:

- 1 Export the composition as a macro in Fusion, by right clicking on a node and selecting Create Macro from the contextual menu.
- 2 Save the macro file (.setting) to the following directory:

MacOS: /Library/Application Support/Blackmagic Design/DaVinci Resolve/Fusion/Templates/Edit/Effects

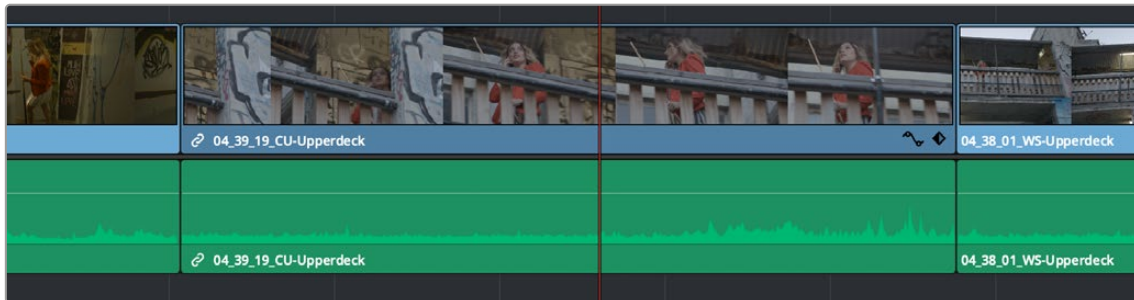
Windows: C:\ProgramData\Blackmagic Design\DaVinci Resolve\Fusion\Templates\Edit\Effects

Linux: /opt/resolve/Fusion/Templates/Edit/Effects

The effect will now be available as a drag and drop effect in the “Fusion Effects” section of the toolbox in the Effects Library on the Edit page.

Seeing Effects in the Timeline

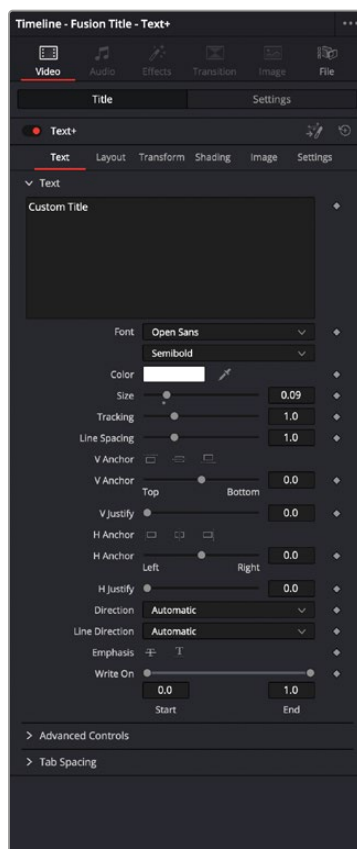
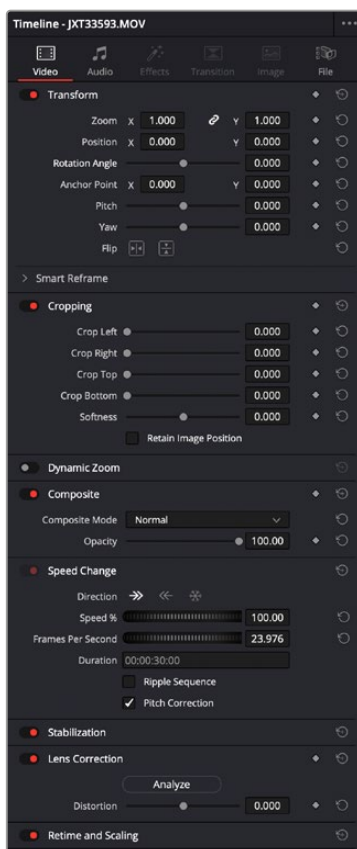
When you apply any kind of an effect to a clip in the Timeline, be it an adjustment in the Inspector, a speed effect, a plugin you've applied, etc., clips with that effect appear a darker shade of whatever color they are to show you there's an effect applied. Removing all effects from a clip (for example, using the Remove Attributes command) returns that clip to its original color. This makes it easy to see, at a glance, which clips have effects, and which clips don't.



A clip with effects that's shaded darker between two other regularly-shaded clips without effects

Using the Inspector

Once you've added effects to a timeline, the Effects Inspector is where you can edit their parameters. The Inspector is the central area for editing all of the settings relating to filters, compositing, sizing, titling, transitions, generators, and effects of all kinds.



The Inspector displays different parameters depending on what you've selected in the Timeline; (left) parameters of a clip, (right) parameters of a title.

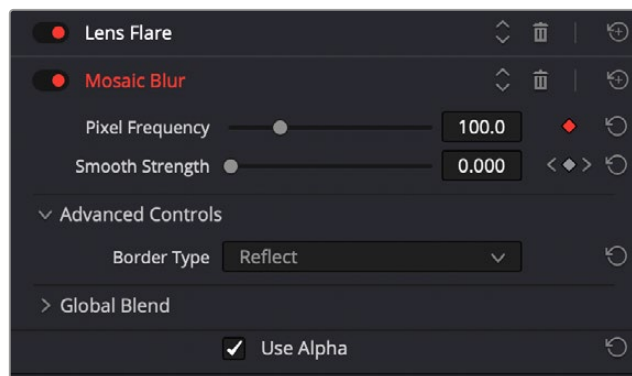
Many of the instructions in this section require the use of the Inspector, which can be opened or closed by clicking the Inspector button at the far right of the Edit and Cut page toolbars, or by double-clicking a transition or generator in the Timeline.

There are three ways that the parameters of clips in the Timeline can be displayed in the Inspector:

- If no clips are selected in the Timeline, then the clip in the highest auto-select-enabled track that intersects the playhead will have its parameters shown in the Inspector.
- The Inspector always shows the parameters of one or more selected items in the Timeline, which will override the clip in the highest track that intersects the playhead, if necessary. Changing the selection changes which parameters are displayed, and the parameters you edit in the Inspector only alter the currently selected clip. If multiple clips are selected, the Inspector displays “Multiple Clips” and allows you to adjust the parameters of all selected clips at the same time.
- Choosing Timeline > Selection Follows Playhead sets DaVinci Resolve to always select whichever clip intersects the playhead in the Timeline. The result is that the Inspector always displays the parameters of the clip at the playhead, with the added bonus that the clip at the playhead is also selected for other editorial functions. If there are multiple superimposed clips intersecting the playhead all at once, the topmost video clip with an enabled Auto Select control will be selected, thus exposing its parameters in the Inspector, and all other clips will be ignored.

Inspector Effects Controls

Different Effects clips in the Timeline expose different controls. Whichever panels are exposed, parameters within each panel are organized into groups, with a title bar providing the name of that group, along with other controls that let you control all parameters within that group at the same time.



The Effects Inspector controls

These controls include:

Enable button: A toggle control to the left of the parameter group's name lets you disable and re-enable every parameter within that group at once. Orange means that track's enabled. Gray is disabled.

Parameter group title bar: Double-clicking the title bar of any group of parameters collapses or opens them. Even more exciting than that, Option-double-clicking the title bar of one parameter group collapses or opens all parameter groups at once.

Keyframe and Next/Previous Keyframe buttons: This button lets you add or remove keyframes at the position of the playhead to or from every single parameter within the group. When the button is highlighted orange, a keyframe is at the current position of the playhead.

When it's dark gray, there is no keyframe. Left and right arrow buttons let you jump the playhead from keyframe to keyframe for further adjustment.

Reset button: Lets you reset all parameters within that group to their default settings.

Adding Filters to Video Clips

DaVinci Resolve supports both built-in Resolve FX and third-party OFX plugins to create various effects. These effects can be applied both to clips in the Edit page, and to nodes in the Color page. This section shows how to apply, edit, and remove these filters in the Edit page.

For more information about using video effects in the Color page, see *Chapter 151, "Using Open FX and Resolve FX."*

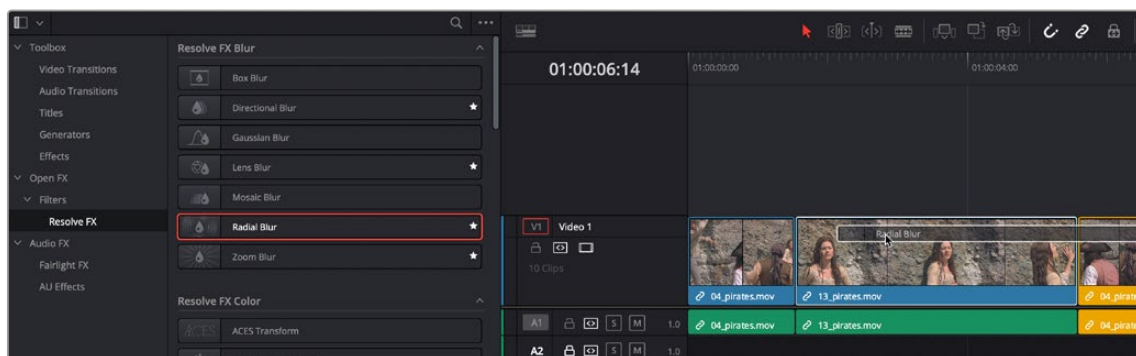
For a detailed explanation of each of the Resolve FX plugins that accompany DaVinci Resolve, see *Part 12, "Resolve FX Overview."*

Methods of applying video filters in the Edit page:

To apply a video filter to a clip:

- Drag any filter from the Effects Library onto the clip in the Timeline you want to apply it to.
- Double Click the filter in the Effects Library to apply it to the selected clip.
- Drag the Filter from the Effects Library to the Inspector to apply it to the selected clip.
- Drag the Filter from the Effects Library to the Viewer to apply it to the clip being viewed.

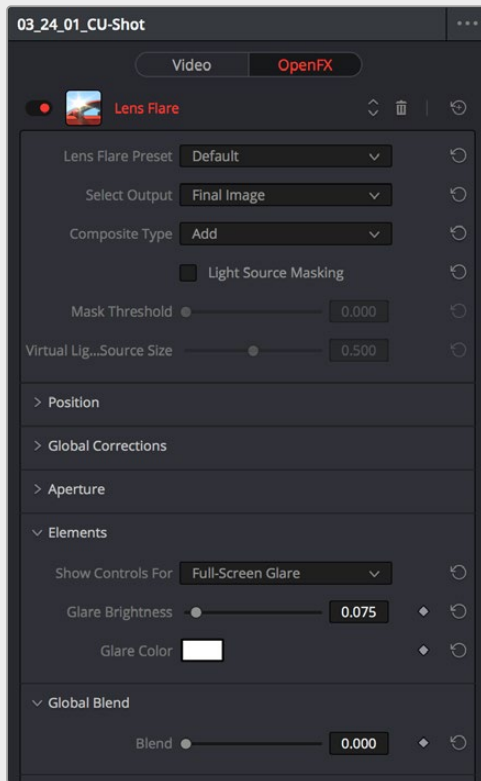
To apply a video filter to multiple clips: Select all of the clips you want to apply a filter to in the Timeline, and then drag any filter from the Open FX category of the Effects Library onto any of the selected clips. This is undoable.



Applying a video filter to a single clip in the Timeline

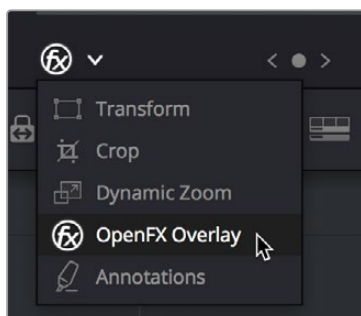
To edit a clip's video filters:

Select that clip and open the Inspector's Effects tab. The effects will be further sub-grouped by type: Fusion, Open FX, or Audio. If a clip does not have any effects assigned to it, this panel will be dimmed.



Resolve FX controls appear in the Effects panel of the Inspector.

Some video filters have custom onscreen controls that can be modified in the Viewer. These can be exposed in the Edit page using the OFX mode of the Viewer.



Turning on the onscreen controls for Resolve FX in the Edit page Timeline Viewer

Once enabled, the OFX onscreen controls appear in the Viewer.



Modifying onscreen controls for Resolve FX in the Edit page Timeline Viewer

Many audio filters expose custom controls that appear in a floating window.

To expose a filter's custom controls:

Open the parameters if they're not open already by double-clicking that filter's title bar. A button should appear at the top of the parameters for filters that have custom UI. Clicking this button opens a floating window with all the custom controls. When you're finished adjusting the custom controls, close the window.



The Fairlight FX Noise Reduction custom UI interface

Methods of working with video filters in the Inspector:

- **To rearrange the order of multiple video filters applied to a clip:** Click the move up or move down buttons in any filter's title bar, to the left of each filter's Trash Can button.
- **To disable or re-enable a filter:** Click the toggle control at the far left of each filter's title bar. Orange means that track's enabled. Gray is disabled.
- **To remove a filter:** Click the Trash Can button.
- **To reset a filter:** Click the Reset button at the far right of the filter's title bar.
- **To open or collapse a filter's parameters:** Double-click the title bar.
- **To open or collapse the parameters of all filters:** Option-click the title bar.

Once applied to a clip, video filters can also be keyframed or automated just like any other Inspector setting, to create dynamic effects that change over time.

Render in Place

Render in Place allows you to render and bake in all effects that are applied to a single clip on the Edit page Timeline. This command, which only works in the Edit page, creates an entirely new media file that replaces the original clip on the Timeline. This new file is created in the same directory as the original source file and is added to the Media Pool automatically.

You can use Render in Place to improve the playback performance of a computationally intensive clip, or use it to create a new high-quality master media with effects that have been finalized baked in. For example, perhaps you have created a clip with a complicated speed ramp, and you want to pass it to another editor or vfx artist in a round-trip rendering scenario, but you are worried about how other programs may interpret the speed effects. In this scenario you could render the clip in place at master quality, and then render and deliver the program.

Render in Place is not a one way operation. Afterwards, you have the option to "Decompose to Original" to bring back the original clip with the original effects, if you need to make a change because the clip was not really as finalized as you were hoping.

To Render in Place:

- 1 Select one or more clips on the Edit Page timeline. Selecting multiple clips results in each clip being individually rendered in place, but as a batch operation.
- 2 Right-click the selection, and choose "Render in Place" from the contextual menu.
- 3 Choose the appropriate Render Clip Options, and then click the "Render" button.

Start Timecode: Sets the starting timecode value for the clip.

Format: Selects the media file format.

Codec: Selects the video codec.

Type: Specifies the compression parameters of the selected codec.

Include Handles: Gives you the option to specify the number of frames before and after the clip In/Out points to be rendered.

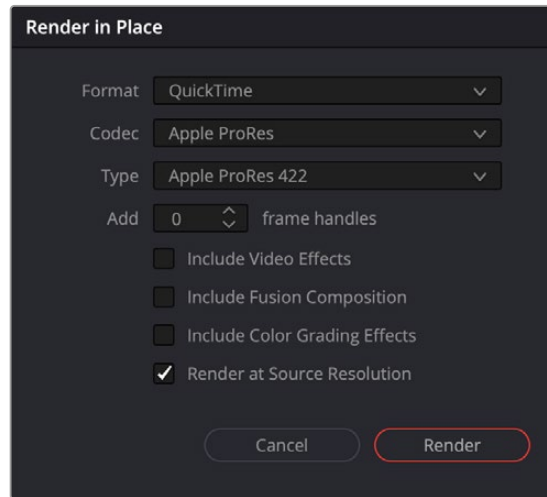
Include Video Effects: Turn on this checkbox to bake in all effects that have been applied to the clip, such as sizing, Open FX or Resolve FX, and speed effects. Turning this checkbox off renders the clip with speed effects baked in, but no other effects applied.

Include Fusion Composition: Turn on this checkbox to bake in any compositions attached to the clip.

Include Color Grading Effects: Turn on this checkbox to bake in any color grading attached to the clip from the Color page.

Render at Source Resolution: When checked the newly rendered media will be at the resolution of the original source media, rather than the timeline resolution.

- 4 Use the File dialog that appears to choose where you want to save the resulting media. Choose a location and click Open.



The Render in Place options

A progress bar appears to show you how long this will take. When finished, your new media is saved in the designated location, added to the Media Pool, and will replace each corresponding source clip on the Timeline.

If you've used Render in Place and end up having buyer's remorse, or a late-breaking change comes back to you later, you can easily decompose to the original clip with its editable effects to make the change.

To Decompose to Original:

- 1 Select one or more clips that have already been Rendered in Place, on the Edit Page timeline.
- 2 Right-click the selection, and choose "Decompose to Original" from the contextual menu.

The original clip, along with all of its editable effects, will be returned to the Timeline. The new media created in the Render in Place process will not be deleted from the source folder, nor will it be removed from the Media Pool. It is effectively a new clip.

Fusion Effect Templates Cache

Fusion effect templates applied to clips on the timeline have more granular caching as well as automatic render caching options.

The Render Cache Fusion Effect Filter menu in the timeline clip context menu has an Auto option that is enabled by default. When render cache is set to Smart, any Fusion effects and templates applied to a clip are automatically cached. Disabling auto allows you to individually pick which effects to cache.

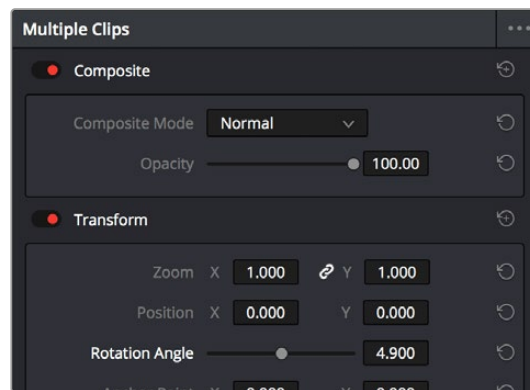
When render cache is set to User, the Auto option automatically caches the Fusion effects, if the project setting to Automatically Cache Fusion Effects in User mode is enabled. If the project setting is disabled, Auto mode does not cache any Fusion effects, unless you individually select them.

Adjusting Multiple Clips at the Same Time

There's an easy way to make adjustments to the Inspector parameters of multiple clips at the same time, without needing to use Paste Attributes (described later in this chapter). All you need to do is simultaneously select every clip you want to alter, and then modify the parameter in the Inspector that you want to change. As a result, every selected clip will be adjusted by the same amount. This works for compositing effects, transforms, text parameters, filters, and audio settings, just about anything that can be simultaneously exposed in the Inspector for multiple selected clips.

When you select multiple clips, the Inspector will display "Multiple Clips" as the title. If each of the selected clips have different values in the parameter you're adjusting, that parameter will have two dashes in the value field. There are two ways you can make adjustments to multiple clips:

- If you want to make a relative adjustment to all selected clips while keeping their original offsets from one another, then drag the virtual slider in the parameter field which will display a + or – before however many units your adjustment is.
- However, if you want to set all selected clips to the same value, you can double-click in the number field, type the value, and press Return.



Making a relative adjustment of plus 4.9 in the Rotation Angle of all selected clips

Adjustment Clips

You can also apply all sorts of effects to multiple clips in the Timeline using Adjustment clips, available from the Effects bin of the Toolbox in the Effects Library. When an Adjustment Clip is superimposed above one or more clips in the Timeline, any filters or other effects that are applied to the Adjustment clip are also applied to all clips underneath it.

Adjustment clips can be used to apply the following types of effects:

- Resolve FX and Open FX plugins
- Inspector parameters including Composite, Transform, Cropping, and Dynamic Zoom
- Fusion page effects
- Color page grading and sizing

Adjustment clips are a fast and easily revised way to apply one or more effects and grades to a range of clips. Adjustment clips that are difficult to playback in real time can be rendered to cache, just like any normal video clip. Adjustment Clips can be named using the Inspector. To store an Adjustment clip, simply drag it from the Timeline to the Media Pool. You can then manage the Adjustment clip just like any other media type.



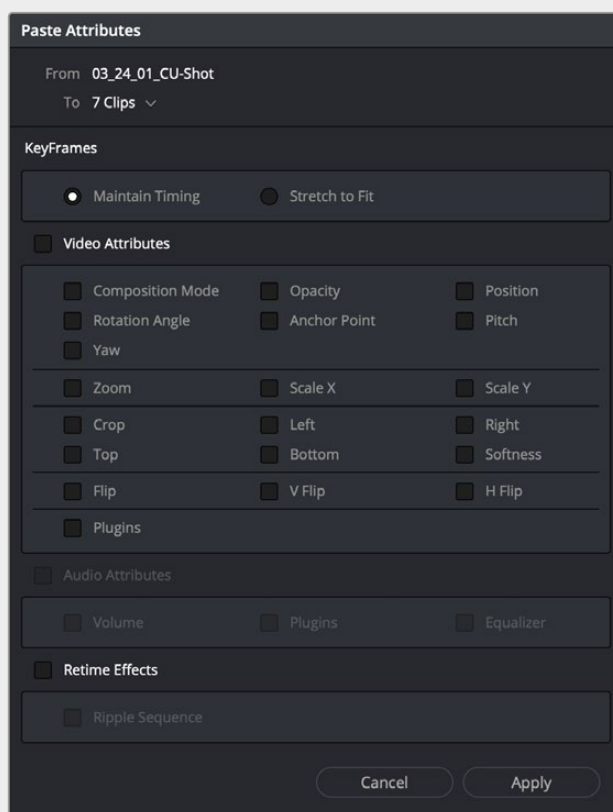
An Adjustment clip applying a Prism Blur Resolve FX's vignette to two other clips

Paste Attributes

You can copy and paste video and audio attributes, as well as color corrections, from one clip to multiple clips using the Paste Attributes command. This is a fast way to apply video and audio adjustments and effects from one clip to many others in the Timeline.

To copy attributes:

- 1 Select a clip with attributes you want to apply to other clips, and press Command-C.
- 2 Select one or more other clips to paste to.
- 3 Choose Edit > Paste Attributes (Option-V), or right-click one of the selected clips and choose Paste Attributes from the contextual menu.
- 4 When the Paste Attributes window appears, click the checkboxes of each of the attributes you want to paste, and click Apply when you're done.



The Paste Attributes window

The Paste Attributes window shows you the clip you're copying from and the clip(s) you're pasting to at the top, and provides checkboxes you can use to select which attributes you'd like to paste.

Keyframe Options for Pasting Keyframed Attributes

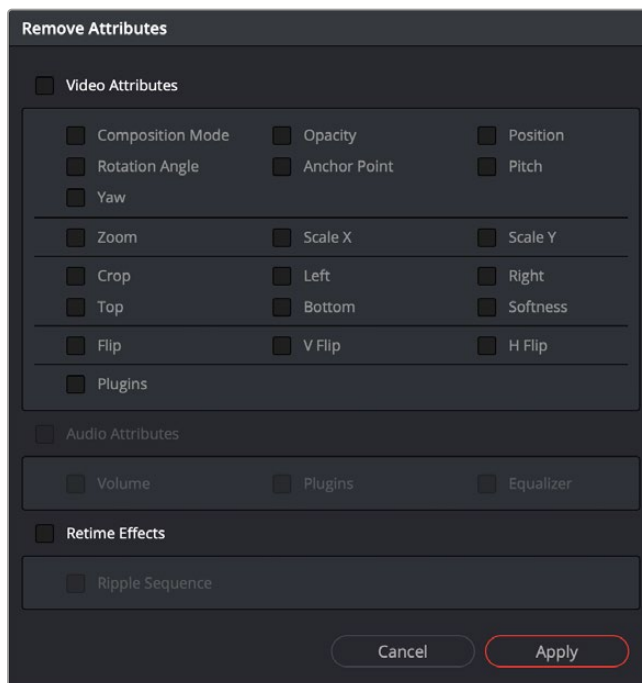
A pop-up menu below lets you choose how you'd like to apply any keyframes that are part of the attributes being pasted; the options are Maintain Timing or Stretch to Fit.

Option to Ripple the Timeline for Pasting Speed Effects

When using Paste Attributes to copy speed effects from one clip to another, the Ripple Sequence checkbox lets you choose whether or not the pasted speed effect will ripple the Timeline.

Remove Attributes

You can also eliminate specific attributes from one or more clips, using a window that's the opposite of the Paste Attributes window.



The Remove Attributes window

To remove attributes:

- 1 Select one or more other clips that have effects you want to remove.
- 2 Choose Edit > Remove Attributes, or right-click one of the selected clips and choose Remove Attributes from the contextual menu.
- 3 When the Remove Attributes window appears, the checkboxes of effects that can be removed are enabled, so turn on attributes you want to remove.
- 4 (Optional) If one of the attributes you're removing is a retime effect, use the Ripple Sequence checkbox to choose whether or not the Timeline will ripple as a result.
- 5 Click Apply when you're done.